RAPIDO RIDE ANALYTICS & PERFORMANCE REPORT

1. Executive Summary

This report presents an end-to-end analytical evaluation of ride-hailing operations using SQL and Power BI. The primary objective of this project is to identify operational inefficiencies, reduce ride cancellations, and optimize revenue through data-driven insights.

The analysis is based on a high-volume dataset consisting of over **1.03 lakh rides**, generating approximately **₹35 million in total revenue**, with **28.9K ride cancellations** and **1.46 million kilometers traveled**.

The findings reveal that while overall service quality remains stable with an average rating of 4.0, a significant proportion of cancellations—particularly from drivers—indicates operational inefficiencies. The dashboard enables real-time monitoring of ride performance, helping stakeholders make faster and more informed decisions.

2. Business Context & Problem Statement

The ride-hailing industry operates in a highly dynamic environment where demand, supply, and service quality must be managed efficiently. However, the organization faced several operational challenges due to the absence of a centralized analytical system.

Key challenges include:

- High ride cancellation rates impacting revenue and customer satisfaction
- Lack of visibility into driver performance and behavior
- Uneven demand distribution across locations
- Inefficient allocation of vehicles across service categories
- Limited insights into payment trends and customer preferences

These challenges resulted in inefficiencies in operations and reduced overall service effectiveness.

3. Objectives

The key objectives of this project are:

- Analyze ride booking trends and operational performance
- Identify root causes of ride cancellations
- Evaluate revenue distribution across locations and vehicle types
- Assess customer and driver satisfaction
- Develop an interactive dashboard for real-time decision-making

4. Data Architecture & Methodology

A structured analytical framework was implemented using a layered approach:

Data Layer (SQL)

The dataset was stored and processed in a MySQL database. SQL queries and views were created to clean, transform, and aggregate the data for analysis.

Semantic Layer (DAX)

DAX measures were used in Power BI to compute key performance indicators dynamically.

Visualization Layer (Power BI)

An interactive dashboard was developed to visualize ride trends, cancellations, revenue, and ratings.

5. SQL Analysis

SQL was used to create structured views and extract meaningful insights from raw booking data. These queries simplified complex business logic and improved data accessibility.

Key SQL Queries:

#1. Retrieve all successful bookings:

```
Create View Successful_Bookings As
SELECT * FROM bookings
WHERE Booking_Status = 'Success';
```

#2. Find the average ride distance for each vehicle type:

```
Create View ride_distance_for_each_vehicle As
SELECT Vehicle_Type, AVG(Ride_Distance)
as avg_distance FROM bookings
GROUP BY Vehicle_Type;
```

#3. Get the total number of cancelled rides by customers:

```
Create View cancelled_rides_by_customers As
SELECT COUNT(*) FROM bookings
WHERE Booking_Status = 'cancelled by Customer';
```

#4. List the top 5 customers who booked the highest number of rides:

```
Create View Top_5_Customers As
SELECT Customer_ID, COUNT(Booking_ID) as total_rides
FROM bookings
GROUP BY Customer_ID
ORDER BY total_rides DESC LIMIT 5;
```

#5. Get the number of rides cancelled by drivers due to personal and car-related issues:

```
Create View Rides_cancelled_by_Drivers_P_C_Issues As
SELECT COUNT(*) FROM bookings
WHERE Canceled_Rides_by_Driver = 'Personal & Car related issue';
```

#6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

```
Create View Max_Min_Driver_Rating As
SELECT MAX(Driver_Ratings) as max_rating,
MIN(Driver_Ratings) as min_rating
```

```
FROM bookings WHERE Vehicle_Type = 'Prime Sedan';
```

#7. Retrieve all rides where payment was made using UPI:

```
Create View UPI_Payment As
SELECT * FROM bookings
WHERE Payment_Method = 'UPI';
```

#8. Find the average customer rating per vehicle type:

```
Create View AVG_Cust_Rating As
SELECT Vehicle_Type, AVG(Customer_Rating) as avg_customer_rating
FROM bookings
GROUP BY Vehicle_Type;
```

#9. Calculate the total booking value of rides completed successfully:

```
Create View total_successful_ride_value As
SELECT SUM(Booking_Value) as total_successful_ride_value
FROM bookings
WHERE Booking_Status = 'Success';
```

#10. List all incomplete rides along with the reason:

```
Create View Incomplete_Rides_Reason As
SELECT Booking_ID, Incomplete_Rides_Reason
FROM bookings
WHERE Incomplete_Rides = 'Yes';
```

#1. Retrieve all successful bookings:

```
Select * From Successful_Bookings;
```

#2. Find the average ride distance for each vehicle type:

```
Select * from ride_distance_for_each_vehicle;
```

#3. Get the total number of cancelled rides by customers:

```
Select * from cancelled_rides_by_customers;
```

#4. List the top 5 customers who booked the highest number of rides:

```
Select * from Top_5_Customers;
```

#5. Get the number of rides cancelled by drivers due to personal and car-related issues:

```
Select * from Rides_cancelled_by_Drivers_P_C_Issues;
```

#6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

Select * from Max_Min_Driver_Rating;

#7. Retrieve all rides where payment was made using UPI:

Select * from UPI_Payment;

#8. Find the average customer rating per vehicle type:

Select * from AVG_Cust_Rating;

#9. Calculate the total booking value of rides completed successfully:

Select * from total_successful_ride_value;

#10. List all incomplete rides along with the reason:

Select * from Incomplete_Rides_Reason;

These queries helped in identifying high-value customers, tracking successful rides, and calculating revenue.

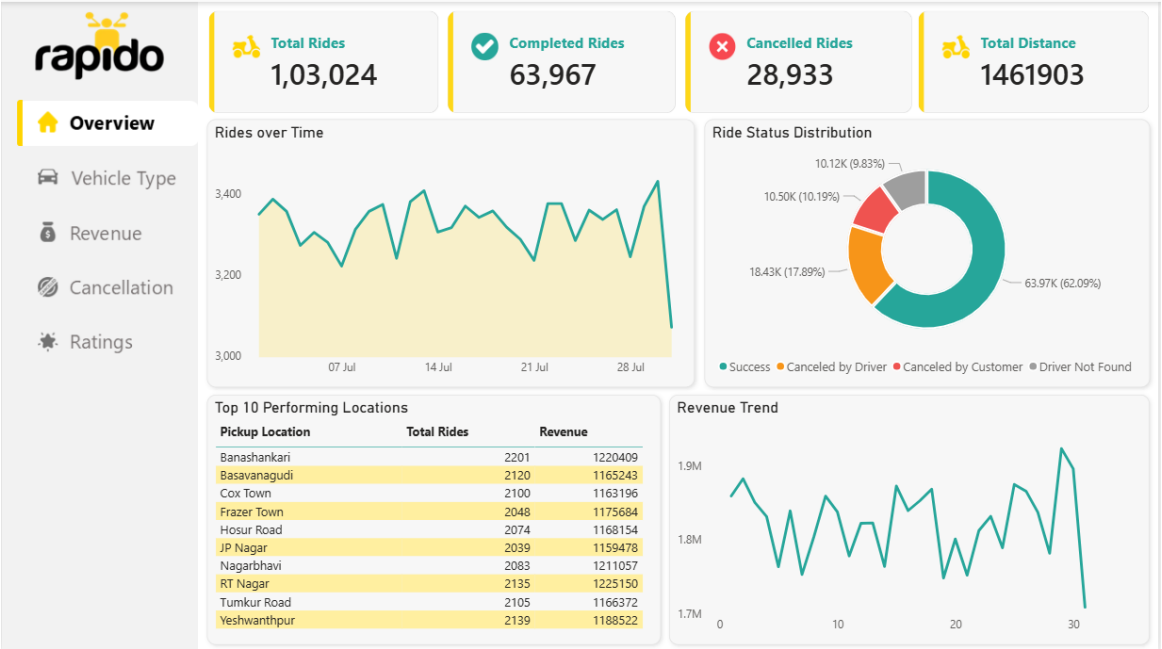
6. Dashboard Overview

The Power BI dashboard provides a comprehensive view of ride performance across multiple dimensions. It is structured into key analytical modules:

- **Overview:** Total rides, completed rides, cancellations, and distance traveled
- **Vehicle Analysis:** Performance comparison across Cab, Auto, Bike, and eBike
- **Revenue Analysis:** Revenue by payment method, vehicle type, and location
- **Cancellation Analysis:** Breakdown of cancellations by drivers and customers
- **Ratings Analysis:** Evaluation of customer and driver satisfaction

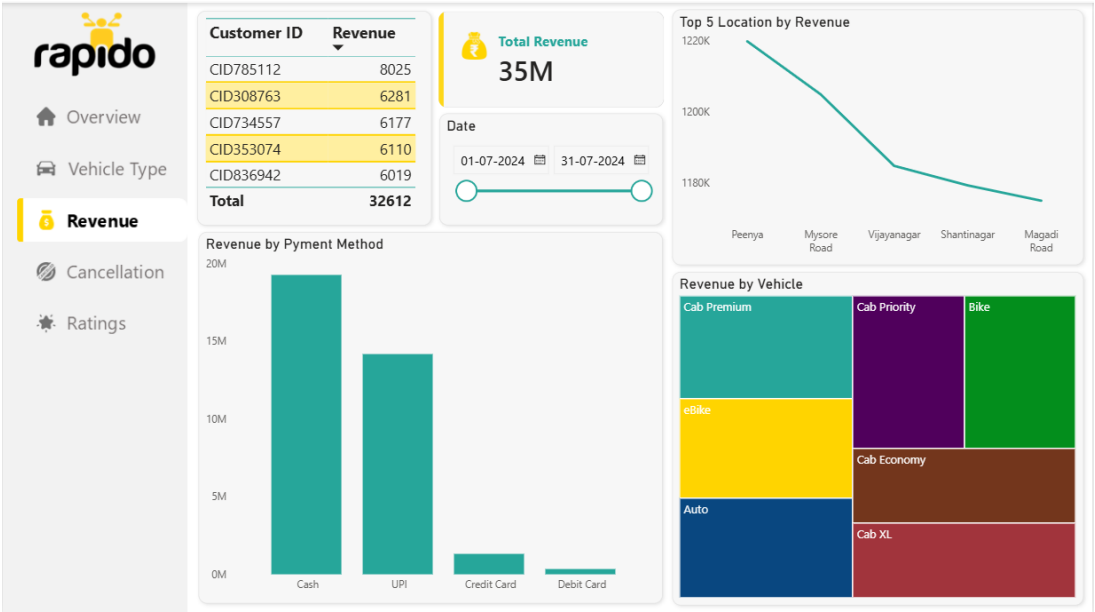
7. Dashboard Visuals

- Dashboard Overview



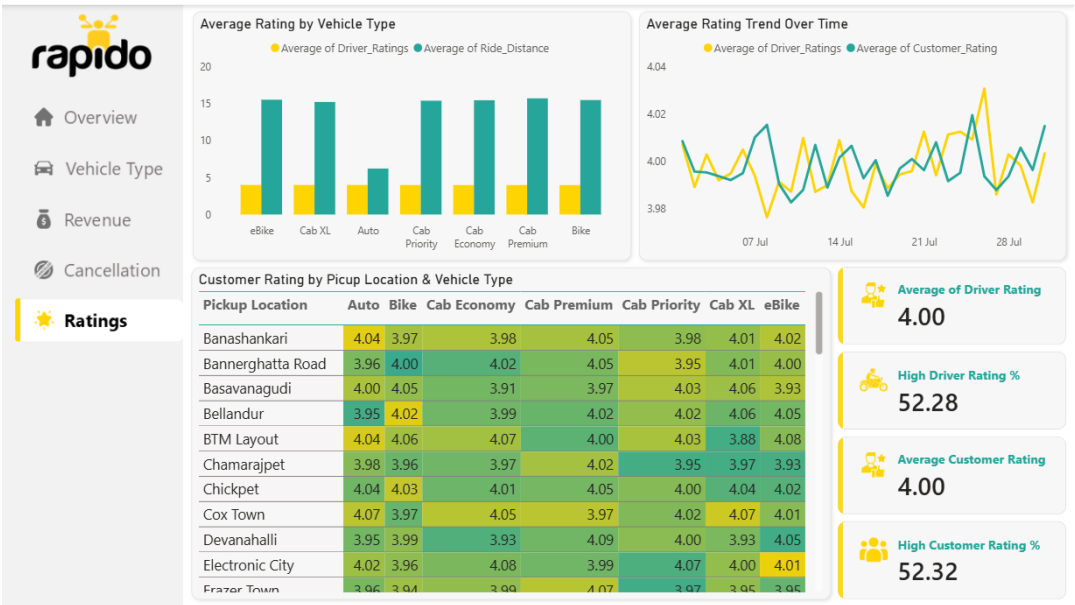
Pickup Location	Total Rides	Revenue
Banashankari	2201	1220409
Basavanagudi	2120	1165243
Cox Town	2100	1163196
Frazer Town	2048	1175684
Hosur Road	2074	1168154
JP Nagar	2039	1159478
Nagarbhavi	2083	1211057
RT Nagar	2135	1225150
Tumkur Road	2105	1166372
Yeshwanthpur	2139	1188522

- Revenue Analysis



- Cancellation Analysis

- Ratings Analysis



Pickup Location	Auto	Bike	Cab Economy	Cab Premium	Cab Priority	Cab XL	eBike
Banashankari	4.04	3.97	3.98	4.05	3.98	4.01	4.02
Bannerghatta Road	3.96	4.00	4.02	4.05	3.95	4.01	4.00
Basavanagudi	4.00	4.05	3.91	3.97	4.03	4.06	3.93
Bellandur	3.95	4.02	3.99	4.02	4.02	4.06	4.05
BTM Layout	4.04	4.06	4.07	4.00	4.03	3.88	4.08
Chamarajpet	3.98	3.96	3.97	4.02	3.95	3.97	3.93
Chickpet	4.04	4.03	4.01	4.05	4.00	4.04	4.02
Cox Town	4.07	3.97	4.05	3.97	4.02	4.07	4.01
Devanahalli	3.95	3.99	3.93	4.09	4.00	3.93	4.05
Electronic City	4.02	3.96	4.08	3.99	4.07	4.00	4.01
Frazer Town	3.96	3.94	3.99	4.07	3.97	3.95	3.95

8. DAX Measures

DAX measures were implemented to calculate key metrics and enable dynamic dashboard interaction.

```
-- #1. Total Volume Tracking
Total Rides = COUNT(Bookings[Booking_ID])

-- #2. Aggregate Cancellation Metrics
Total Cancelled Rides =
CALCULATE(
   COUNT(Bookings[Booking_ID]),
   Bookings[Booking_Status] IN { "Canceled by Driver", "Canceled by Customer" }
)

-- #3. Quality of Service (Ratings Analysis)
High Driver Rating % =
DIVIDE(
   CALCULATE(COUNT(Bookings[Driver_Ratings]), Bookings[Driver_Ratings] >= 4),
   COUNT(Bookings[Driver_Ratings])
) * 100

High Customer Rating % =
DIVIDE(
   CALCULATE(COUNT(Bookings[Customer_Rating]), Bookings[Customer_Rating] >= 4),
   COUNT(Bookings[Customer_Rating])
) * 100

-- #4. Driver vs Customer Cancellation Impact
Cancelled by Driver =
CALCULATE(COUNT(Bookings[Booking_ID]), Bookings[Booking_Status] = "Canceled by Driver")

Driver Cancel % = DIVIDE([Cancelled by Driver], [Total Rides], 0)

Cancelled by Customer =
CALCULATE(COUNT(Bookings[Booking_ID]), Bookings[Booking_Status] = "Canceled by Customer")

Customer Cancel % = DIVIDE([Cancelled by Customer], [Total Rides], 0)
```

These measures enable real-time analysis and improve dashboard responsiveness.

9. Key Insights & Findings

The analysis reveals several critical insights:

- The ride success rate is approximately **62%**, indicating scope for improvement
- Driver cancellations (**18.4K**) are significantly higher than customer cancellations (**10.5K**)
- Cash remains the dominant payment method, though digital payments are increasing
- Certain locations contribute significantly more revenue, indicating demand concentration
- Vehicle categories such as Cab Premium and Bike are key revenue drivers
- Customer ratings remain consistently high, indicating satisfactory service quality

10. Business Impact

The implementation of this dashboard significantly improves the organization's analytical capabilities. It enables real-time monitoring, reduces reliance on manual reporting, and enhances decision-making.

The solution helps identify inefficiencies, optimize operations, and improve customer experience, ultimately contributing to better business performance.

11. Strategic Recommendations

Based on the analysis, the following recommendations are proposed:

- Introduce incentive programs to reduce driver cancellations
- Improve driver allocation in high-demand areas
- Promote digital payments such as UPI
- Optimize pricing strategies based on demand patterns
- Address operational issues leading to cancellations
- Enhance service quality to maintain high ratings

12. Future Scope

Future enhancements include:

- Implementation of demand forecasting models
- Integration of real-time ride data
- Advanced customer segmentation
- Deployment of dashboard on cloud platforms

13. Conclusion

This project demonstrates how data analytics and visualization can transform raw operational data into actionable insights. The integration of SQL, DAX, and Power BI provides a scalable and efficient analytical solution for ride-hailing operations.

The insights generated from this analysis can significantly improve operational efficiency, reduce cancellations, and enhance customer satisfaction.